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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S, FE) / S4 (PT) (S,FE) / S2 (PT) (S,FE) / S4 (WP) (S) Examination December 2024 (2019 Scheme)

Course Code: MAT204**Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 The probability that there will be 0,1,2,3 defective light bulbs in a randomly selected batch of bulbs produced in a factory are 0.6, 0.3, 0.08, 0.02 respectively. Find the mean and variance of the number of defective bulbs in a randomly selected batch. (3)
- 2 If X is a Poisson random variable with $E[X] = 6$, find $P(X = 4)$. (3)
- 3 Consider a continuous random variable X with the probability density function $f(x) = 2x$, $0 \leq x \leq 1$. Find $P(0.25 \leq X \leq 0.75)$ and $E[X]$. (3)
- 4 For an exponential random variable X , given that $P(X > 2) = e^{-10}$. Find $P(X > 7/X > 5)$. (3)
- 5 Find the variance at time $t = 3$ of a WSS random process $X(t)$ with autocorrelation function $R_X(\tau) = e^{-|\tau|}$. (3)
- 6 Show that the power spectral density of a WSS process is a real valued even function. (3)
- 7 Find the first iterate x_1 for the root of the function $f(x) = x^3 - 5x - 9$ using Newton-Raphson method with initial guess $x_0 = 2$. (3)
- 8 Find the divided differences for the following data points (1,4), (2,7), (3,12), (4,19). (3)
- 9 Write the normal equations for fitting a parabola of the form $y = a + bx + cx^2$ to a given data by the method of least squares. (3)
- 10 Write any sufficient conditions for the Gauss-Seidel method for iterating the solution of a linear system to converge. (3)

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PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11 a) A factory produces electronic components, and historically, 5% of these components are defective. If a batch contains 200 components: (7)
- (i) What is the probability that exactly 15 components are defective according to the binomial distribution?
- (ii) Using the Poisson approximation to the binomial distribution, estimate the probability that the batch contains at most 5 defective components.
- b) Find the mean and variance of a Poisson random variable with parameter λ . (7)
- 12 a) Let X be a discrete random variable with the following probability mass function (7)
- $P(X = 1) = 0.3, P(X = 2) = 0.2, P(X = 3) = 0.1, P(X = 4) = 0.4$
- (i) Verify that this function satisfies the properties of a PMF.
- (ii) Calculate the probability that X is an even number.
- (iii) Determine the cumulative distribution function (CDF) of X .
- b) Let X and Y be discrete random variables with the following joint probability mass function (PMF): (7)

	$Y = -1$	$Y = 0$	$Y = 1$
$X = 1$	0.1	0.2	0.1
$X = 2$	0.2	0.15	0.25

- (i) Calculate $P(X = 2, Y \leq 0)$
- (ii) Find the marginal PMFs of X and Y .
- (iii) Determine the expected values of X and Y .

Module -2

- 13 a) The heights of a population of adult males in a certain country follow a normal distribution with a mean of 175 cm and a standard deviation of 7 cm. (7)
- (i) What percentage of adult males in this country have a height greater than 185 cm?
- (ii) If 200 adult males are randomly selected from this population, what is the expected number of individuals with a height between 170 cm and 180 cm?

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- b) Passengers come to a bus station at random times. The waiting time of a random passenger, until his bus comes, follows a uniform distribution between 5 and 15 minutes. (7)
- (i) Calculate the probability that a person waiting at this bus stop will wait for more than 10 minutes.
- (ii) If 120 people arrive at this bus stop independently, what is the expected number of people who will wait between 7 and 12 minutes?
- 14 a) Random variables X and Y have joint probability density function (7)
- $$f(x, y) = \begin{cases} k(x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$
- (i) Find the value of k so that $f(x, y)$ is a valid probability function.
- (ii) Find $P\left(0 < X < \frac{1}{4}, 0 < Y < \frac{1}{4}\right)$.
- (iii) Find the marginal density functions of X and Y.
- b) A study on the average monthly internet usage in a small town shows a wide range of internet usage values with a mean of 250 GB and a standard deviation of 60 GB. What is the probability that for a random sample of 100 residents from this population (i) the mean monthly internet usage is 260 GB or more? (7)
- (ii) the total monthly usage is less than 20000 GB .

Module -3

- 15 a) A random process $X(t)$ is defined by $X(t) = A \cos(\omega t + \phi)$ where ω is a constant, A is a random variable which is uniformly distributed in $[0, 1]$ and ϕ is a random variable, independent of A and uniformly distributed in $[0, 2\pi]$. Determine whether the random process $X(t)$ is Weakly Stationary (WSS) or not. (7)
- b) The autocorrelation function of a continuous-time Weakly Stationary (WSS) process $X(t)$ is $R_X(\tau) = \sigma^2 e^{-|\tau|}$, where σ^2 is the variance of the process. Find the power spectral density of the process and the average power of the process. (7)
- 16 a) If people arrive at a book stall in accordance with a Poisson process with mean rate of 3 per minute, find the probability that the interval between 2 consecutive arrivals is (i) more than 1 minute (ii) between 1 minute and 2 minutes (iii) 4 minutes or less. (7)
- b) Prove that autocorrelation of a random process is an even function. (7)

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- 17 a) Solve the equation $xe^x - 2 = 0$ using Regula Falsi Method. (7)
- b) A third degree polynomial passes through the points (0,-1), (1,1), (2,1) and (3,-2). Using Newton's forward interpolation formula find the polynomial. Hence find the value at 1.5
- 18 a) Using Lagrange's interpolation method compute the value of $y(4)$ for the following data (7)

x	1	3	5	7
y	24	120	336	720

- b) Calculate $\int_{0.5}^{0.7} e^{-x} \sqrt{x} dx$ using Simpson's method by taking 5 ordinates. (7)

Module -5

- 19 a) Solve the following system of equations using Jacobi's method (7)
- $$27x + 6y - z = 85, \quad x + y + 54z = 110, \quad 6x + 15y + 2z = 72$$
- b) Using Euler's method, find $y(0.5)$ of $y(x)$ which satisfies the initial value problem, $\frac{dy}{dx} = \frac{1}{2}(x^2 + 1)y^2$, given $y(0.2) = 1.1114$. Take $h = 0.1$. (7)
- 20 a) Using Runge-Kutta method of order 2, evaluate $y(0.1)$ if $\frac{dy}{dx} = x^2 + y^2$, given that $y(0) = 1$. (7)
- b) Using Adams Moulton method find $y(4.4)$ for the differential equation (7)
- $$\frac{dy}{dx} = \frac{2-y^2}{5x}. \quad \text{Given } y(4) = 1, y(4.1) = 1.0049, y(4.2) = 1.0097 \quad \text{and} \\ y(4.3) = 1.0143$$
